

Empirical Evaluation of Breast Cancer Detection using Machine Learning and Deep Learning Techniques

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Abstract— Breast cancer is a prevalent and serious condition affecting women worldwide. Early and accurate diagnosis is critical for determining cancer prognosis and improving patient outcomes. Despite advancements in diagnostic technologies, existing methods often lack accuracy and generalizability, highlighting the need for reliable early detection and classification systems. This study preprocesses breast cancer datasets to ensure clean data, extracts key features, and selects them to enhance model performance. The datasets are divided into training and testing sets, where the training data builds the model, and the testing data validates its accuracy. Cases are classified as normal or abnormal, with abnormal cases further categorized as benign or malignant. Performance is evaluated using metrics such as accuracy, precision, and F1-score. The results demonstrate promising accuracies of 97.81% with 70% training data, 98% with 80%, and an ideal 100% with 90% training data. These findings underscore the effectiveness of the proposed system in improving breast cancer detection and classification, emphasizing the transformative role of artificial intelligence in revolutionizing breast cancer diagnosis and patient care.

Keywords— Artificial Neural Networks (ANN), Convolutional Neural Networks (CNN), K-Nearest Neighbour (KNN), Patch-Based Deep Belief Network for Breast Cancer (Pa-DBN-BC), Support Vector Machine (SVM)

I. INTRODUCTION

Breast cancer is a harmful tumor that swells up in the breast tissue. India recorded the highest number of breast cancer-related deaths (98,337) in 2022 [1]. It normally starts in the surface cells of these ducts or lobules. This illness primarily affects women, though rare cases have been reported in men. Early detection of breast cancer is one of the key factors in getting the best diagnosis and the most efficient treatment. If the problem is not discovered in time, breast cancer can damage the surrounding areas and lymph nodes as well as migrate to bones, lungs, liver, or brain. Since breast cancer is

unavoidable, it makes treatment more challenging and often necessitates aggressive options with increased side effects, which lead to a lower survival rate.

The cell types that are related to breast cancer include: epithelial cells (most breast cancers begin from these cells), myoepithelial cells (which act as tumor suppressors in the early stages but might be involved in invasion in the later stages), stromal cells (which influence tumor growth and progression through signaling molecules), immune cells (which either promote or suppress tumor growth based on their type and activation state) [2], and endothelial cells (which help tumors grow by forming new blood vessels to supply nutritional materials and oxygen) [3]. The main cell type that causes breast cancer is the epithelial cell, as mutations in these cells result in uncontrolled growth leading to tumor formation.

A tumor is a lump consisting of abnormal tissue. Two types of breast cancer tumors are classified: those that have not spread (noncancerous, also termed ‘benign’) and those that have (cancerous/malignant). If a tumor turns out to be benign, the physician typically leaves it as it is, in most cases, without removing it. Although such tumors are mostly non-invasive, sometimes they increase in size, exert pressure on other parts, and cause pain or other problems. In such cases, surgery of the tumor is performed, and the pain and complications are resolved. Malignant tumors are the ones capable of damaging the tissues around them and are therefore cancerous and potentially fatal. If a tumor is suspected to be malignant, the doctor proceeds to take a biopsy to establish the degree of seriousness or aggressive nature of the tumor.

Breast cancer research has witnessed significant progress over the years, especially in the use of machine learning and deep learning for early-stage diagnosis and classification. In [4], three Machine Learning techniques—Logistic Regression, Random Forest Classification, and K-Nearest Neighbors Classification—were utilized on the Wisconsin Breast Cancer

dataset. Principal Component Analysis and Neighborhood Component Analysis were also performed to enhance the accuracy of the models. K-Nearest Neighbors Classification achieved an accuracy of 99.4%, Random Forest Classification achieved 98.8%, and Logistic Regression achieved 98.2%.

In [5], AlexNet and VGGNet were utilized as feature extractors alongside SVM for mammographic image classification, achieving an accuracy of 64.52%. This study provided a foundation for advancing image classification techniques. As discussed in [6], the DCN-Nin architecture was employed on a dataset of 322 mammographic images, obtaining an accuracy of 60.90%. In [7], classifiers like SVM and Naive Bayes were used on the WBC dataset, with SVM achieving the maximum accuracy of 97.13%.

In [8], GoogLeNet achieved 83% accuracy, while AlexNet achieved 80% accuracy in a study involving 50 patients. As discussed in [9], the Deep Convolutional Neural Network (DCNN) was applied to 2,796 mammographic images, concluding with an accuracy rate of 83.74%. In [10], AlexNet was combined with SVM using the DDSM and CBIS-DDSM datasets, achieving an accuracy of 87.2%. As mentioned in [11], VGG16 and ResNet50 algorithms were applied to the IRMA dataset, with VGG16 achieving 94% accuracy and ResNet50 achieving 91.7% accuracy. In [12], a shallow deep Convolutional Neural Network (CNN) model was used, with an accuracy of 76.05% for DCNN and 89.23% for Stacked AutoEncoders (SAE).

II. RELATED WORK

In [13], a method was proposed using Decision Trees, Random Forests, KNN, SVM, and Artificial Neural Network (ANN) on WDBC and BCWD datasets, achieving an accuracy of 92.05%. As discussed in [14], three CNN models were tested on a dataset obtained from Kaggle, with model 3 being the most accurate, achieving a score of 87%. In [15], the ResNet50 architecture was utilized from the UCI Image Dataset, achieving an accuracy of 95.33%. As proposed in [16], a Weighted Particle Swarm Optimization with a CNN model was applied to public datasets, although specific accuracy metrics were not reported. The study focused on optimizing models for better performance. In [17], a Predictive Analytics - Deep Belief Network for Breast Cancer (Pa-DBN-BC) model was introduced for histopathology images, attaining an accuracy of 86%. As discussed in [18], SVM-CNN, VGG16, ResNet50, and InceptionV3 were tested on a Kaggle dataset, with SVM-CNN achieving the highest accuracy of 93.35%. In [19], the Xception model was applied to breast cancer images, attaining an accuracy of 97.22% for malignant and 98.48% for benign cases. As reported in [20], a custom CNN model was developed, achieving the highest accuracy of 100% on 250,780 images, demonstrating unparalleled performance. In [21], a CNN method was employed on various datasets collected by research institutions, but no accuracy was reported due to the diversity of datasets, preventing conclusive results.

As discussed in [22], MobileNetV2 combined with Random Forest was applied to the DDSM dataset, yielding accuracies of 99.4% and 96%, respectively, highlighting the significance of ensemble methods in classification tasks. In [23], several CNN

architectures were analyzed, with accuracies ranging from 87-91% on the CBIS-DDSM dataset and 91-100% on the INBreast dataset, demonstrating variability across architectures and datasets. As examined in [24], multiple models, including Xception and VGG16, were tested on the BreakHis dataset, reporting accuracies between 98.28% and 98.28%, emphasizing the importance of model selection in achieving high accuracy. In [25], InceptionV3 was employed on a self-curated ultrasound image dataset, achieving an accuracy of 92%, contributing to advancements in ultrasound image analysis. As discussed in [26], DMFTNet was applied to a breast cancer dataset based on methylation, obtaining an accuracy of 76.14%, showcasing the role of specialized structures in classification tasks. In [27], CNN with advanced training techniques was implemented across multiple datasets, achieving an overall accuracy of 95.20%, demonstrating the impact of training strategies on model performance.

As examined in [28], Inception-V3, ResNet50, and AlexNet were applied to mammographic images, achieving accuracies of 97.81%, 98%, and 100% for 70%, 80%, and 90% of the dataset, respectively. In [29], various CNN architectures were compared across multiple datasets, with accuracy ranging from 92% to 96%, reinforcing the necessity of thorough model evaluation.

As highlighted in [30], ResNet50 and InceptionV3 were used on thermogram images, achieving an accuracy of 80%, demonstrating the potential of thermography in cancer detection. In [31], UNet and ResNet50 were integrated for breast cancer image segmentation, achieving an F1 score of 97.30%, highlighting the effectiveness of segmentation models in medical imaging. As discussed in [32], ResNet50 and MobileNet were applied to hospital datasets, although accuracy metrics were not provided, emphasizing the applicability of deep learning in different healthcare settings. In [33], CNN and EfficientNetV2B3 were used on WSI images, attaining an accuracy of 96.30%, reinforcing the strength of advanced CNN models in complex image analysis. As reported in [34], the Inception Network was applied to ultrasound image features, achieving a remarkable accuracy of 99.81%, showcasing the potential of feature extraction techniques in improving classification results.

III. DATA COLLECTION

The use of machine learning in the field of breast cancer detection and classification requires the availability of several datasets that are different from each other. In the case of mammographic imaging, in Fig. 1 the CBIS-DDSM dataset [35], which consists of 10,200 images, and the Mini-DDSM set [36], which contains 9,685 images, are the most important in early breast cancer studies. Another example is the Breast Histopathology Images dataset [37], which contains 278,000 H&E stained images from Kaggle and is used for large-scale image analysis studies. There are also some smaller and more specialized datasets, shown in Fig 3 such as the BreakHis dataset [38], which contains 7,909 images of breast cancer, and the IRMA dataset, which includes over 10,000 medical images for specific diagnoses. These collections give researchers the opportunity to train their models on a huge variety of benign and malignant cases, which consequently increases the

precision of the algorithms and their applicability in various diagnostic scenarios.

Along with mammographic and histopathological imaging datasets, other specialized collections of data have contributed to the study of breast cancer from different perspectives. For example, the Affiliated Hospital of Southwest Medical University and the Guangdong Provincial Hospital of Traditional Chinese Medicine [3] have provided different training and test datasets, which have been of great variability for model evaluations. The MIAS [39] and DDSM [40] datasets, which total over 110,000 images, support large-scale training with diverse samples, while smaller datasets, such as the 2,088 thermographic images, offer alternative imaging modalities. Publicly available resources, like the WBC dataset [41][42] with 569 original images and the methylation breast cancer dataset with 1,538 entries, further diversify the research landscape, enabling in-depth exploration of breast cancer's molecular and cellular features. Together, these datasets play a vital role in refining deep learning models for more accurate and comprehensive diagnostic solutions.

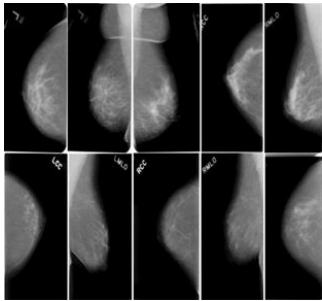


Fig. 1. CBIS-DDSM dataset [35]

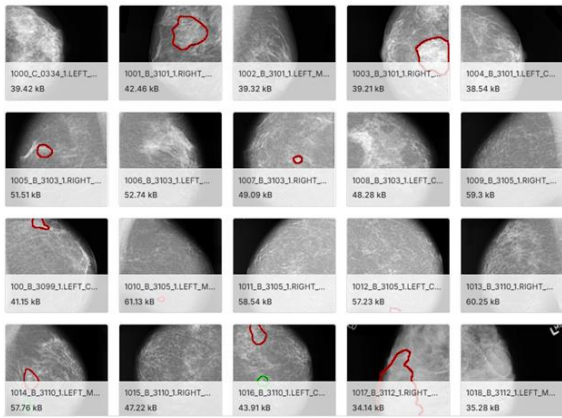


Fig. 2. Mini-DDSM dataset [36]

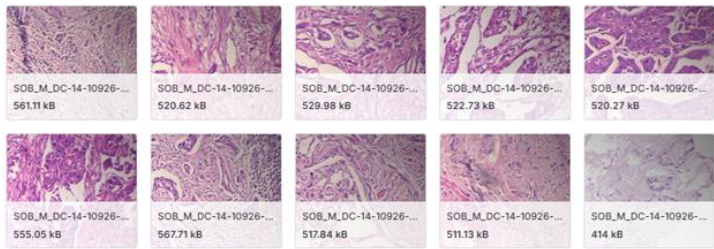


Fig. 3. BreakHis dataset [38]

TABLE I. SUMMARY OF RELATED WORK

S. No.	Year and Technique	Dataset	Accuracy
1.	2015: AlexNet & VGGNet(both as feature extractors) and SVM(as classifies) [5]	600 mammographic images	64.52%
2.	2016: DCN-Nin(efficiency was checked) [6]	322 mammographic images	60.90%
3.	2016: ml methods classifiers were used which includes SVM, Naive Bayes, KNN and decision tree C4.5 [7]	WBC(original) dataset	SVM(97.13%)
4.	2017: GoogLeNet, AlexNe t[8]	50 patients	83%, 80%
5.	2017: DCNN [9]	2796 mammographic images	83.74%
6.	2019: AlexNet & SVM [10]	DDSM, CBIS-DDSM(B & M)	87.2% (CBIS-DDSM)
7.	2019: VGG16, ResNet50 [11]	IRMA Dataset	94%(vgg16), 91.7%(rn)
8.	2019: Shallow Deep Convolutional Neural Network(DCNN), AlexNet, Stacked AutoEncoders(SAE) (all 3 compared) [12]	DDSM(CC and MLO view of breasts) 8000 mammographic images	76.05%, 89.23%, 63.7%
9.	2020: logistic regression, KNN, SVM, Naive Bayes, Decision tree, Random Forest, Rotation forest [13]	WDBC, BCWD, BHI dataset	98.10%
10.	2021: CNN model 1, CNN model 2, CNN model 3 [14]	Kaggle 162 H&E	CNN model 3(87%)
11.	2021: ResNet50[15]	UCI Image Dataset	95.33%
12.	2021: weighted particle swarm optimization (WPSO) & CNN [16]	Publicly available dataset	not reported
13.	2021: Pa-DBN-BC(next for categorization logistic regression) [17]	histopathology images	86%
14.	2022: SVM-CNN, VGG16, RESNET 50, and INCEPTIONV3 [18]	3538 images kaggle dataset	93.35%, 89.54%, 92.45%, 88.6%
15.	2022: xception [19]	breast cancer images	97.22%(mabelignant 98.48%(benign)
16.	2022: CNN custom model(comparison w AlexNet, Darknet19, GoogleNet, MobileNet, ResNet18, ResNet50, VGG16, Xception) each with 3 optimizers SGDM, ADAM, RMSprop [20]	250, 780 images	100.00%
17.	2022: CNN [21]	DDMA, TCIA, other datasets collected by research institutions	not reported

S. No.	Year and Technique	Dataset	Accuracy
18.	2023: MobileNetV2 with Random Forest classifier, seven-layer CNN using scratch learning [22]	DDSM dataset	99.4%, 96%
19.	2023: VGG19, ResNet50, MobileNet-v2, Inception-v3, Xception, and Inception-ResNet-v2 [23]	10,239 CBIS-DDSM and 410 INbreast mammographic images	87–91% CBIS-DDSM, 91–100% INBreast
20.	2023: Xception, InceptionV3, VGG16, MobileNet and ResNet50, BCNN [24]	7909 microscopic images “Cancer Histopathological Image Classification (BreakHis)	97.54%, 95.33%, 98.14%, 97.67%, 93.98%, 98.28%
21.	2023: InceptionV3 [25]	Self-centered Ultrasound Image Dataset	92%
22.	2023: DMFTNet [26]	methylation breast cancer dataset	76.14%
23.	2024: CNN using ReduceLROnPlateau and EarlyStopping [27]	CBIS-DDSM, Mini-DDSM, Breast Histopathology Images	95.20%
24.	2023: Inception-V3, ResNet50, and AlexNet (2 stages, TV algo and MSMV classifier is used) [28]	mammographic image analysis homepage	97.81% for training on 70% of the data, 98% for training on 80% of the data, and meets the ideal value of 100% for training on 90% of the data
25.	2023: AlexNet, ResNet50, EfficientNet, MobileNet, ConvNeXt,Concat. Model [29]	RSNA , MIAS , DDSM	92%, 94.5%, 96%(dataset)
26.	2023: Resnet50, Inception V3, Xception [30]	2088 thermogram images	80%
27.	2023: UNet & ResNet50 [31]	2780 images	F1 97.30%(S), accuracy 98.61%(C)
28.	2024: Resnet50, MobileNet, Inception V3 [32]	Affiliated Hospital of Southwest Medical University (Centre 1, the training set) and Guangdong Provincial Hospital of Traditional Chinese Medicine (Centre 2, the independent test set)	not reported
29.	2024: CNN & EfficientNetV2B3 [33]	2,77,524 WSI images(BC-IDC dataset)	96.30%
30.	2024: Inception Network with LPP feature enhancement [34]	ultrasound images	99.81%

The system’s effectiveness had been assessed using key evaluation metrics, including accuracy, precision, and F1-score. By optimizing these metrics, the system had enhanced early detection and supported timely medical intervention. In Fig. 4, the proposed system for breast cancer detection and classification had followed a systematic approach to ensure accurate predictions.

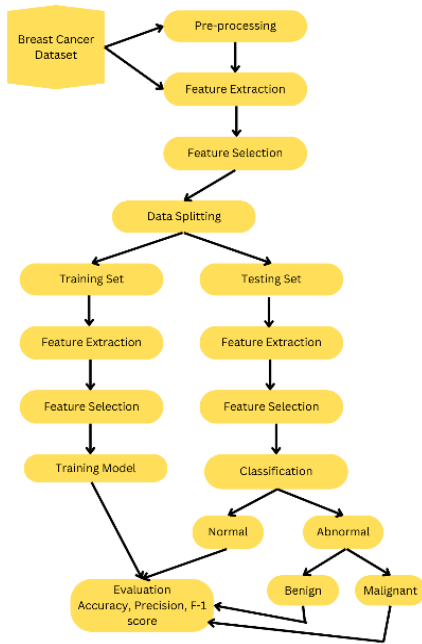


Fig. 4. Proposed System

IV. RESULT AND ANALYSIS

Breast cancer detection techniques have evolved significantly in recent years, embracing a wide array of approaches. This diversity highlights the relentless pursuit of more precise and efficient diagnostic methods. The use of various machine learning and deep learning models underscores the complexity of the challenge and the urgent need for groundbreaking solutions that can make a real difference in people's lives.

A. Techniques Used

Fig. 5 shows the utilization of different approaches like ResNet50, MobileNet, AlexNet, etc. based on the details in Table I.

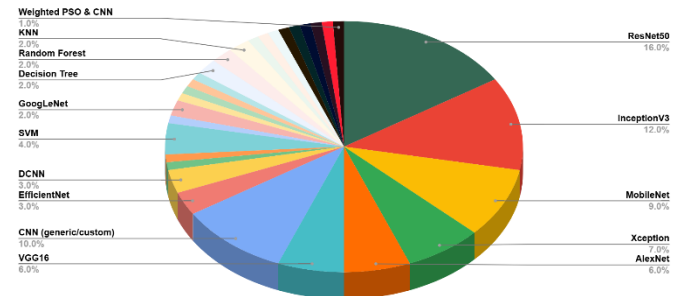


Fig. 5. Techniques Used based on the details in Table 1 (most used technique was ResNet50 used by 16% while least used technique were Weighted PSO & CNN used by 1%)

The segments of the chart had represented various techniques like Weighted Particle Swarm Optimization (PSO)

& CNN, K-Nearest Neighbor (KNN), ResNet50, and others, each with their respective percentages. ResNet50, InceptionV3, and CNN (custom/generic) had been widely used, while KNN, Random Forest, GoogleNet, and DCNN had not been frequently used.

B. No. of publications per year

Fig. 6 illustrates the amount of data that had been collected annually from 2017 to 2024. The y-axis had shown "Count," and the x-axis had shown "Year." The greatest number of research papers on breast cancer detection had been published in 2023.

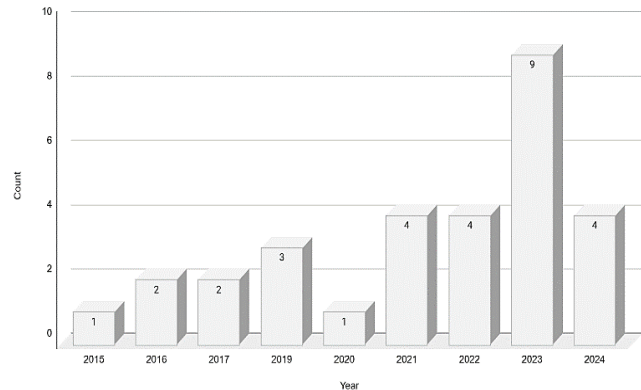


Fig. 6. No. of publications per year based on the details in table 1 (most no of publications were made in 2023 and the least nopublications were made in 2015 & 2020)

C. Data size used in each research

Fig. 7 represents dataset sizes based on citation statistics, such as Muhammet Faith Alk and R. MT. Thakur A Gupta. The percentages had represented the actual proportion of the citations, so data based on the dataset sizes that had been observed was again shown, such as 117% and 105% on some items.

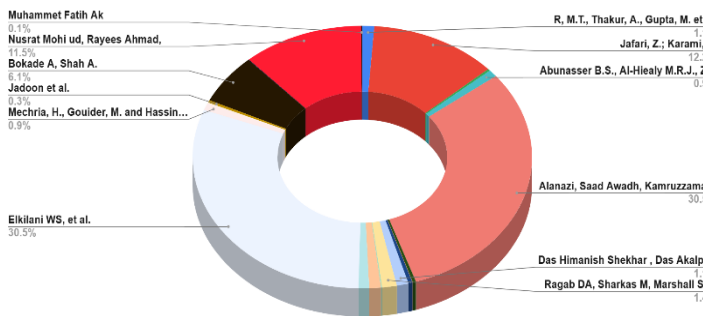


Fig. 7. Data Size used in each research (largest datasize was used by Elkilani,et al.[33])

D. Techniques using Deep Learning models

Fig. 8 compares the accuracy of different research papers using AlexNet. Each research paper had used either different or similar datasets, with specific values indicated on the vertical axis for accuracy. The chart had displayed figures indicative of the different degrees of accuracy that each research paper had achieved. The highest accuracy for AlexNet by using mammographic images was 95.82%.

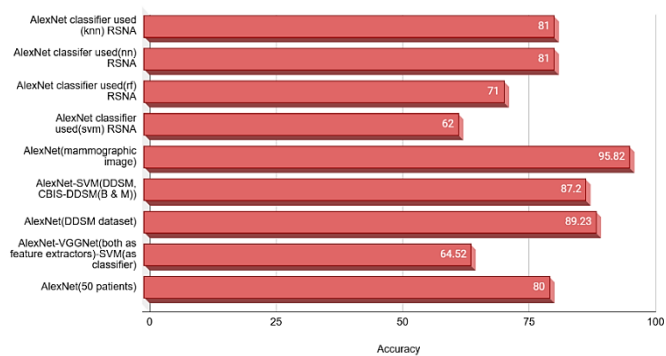


Fig. 8. Accuracy Vs Techniques using AlexNet

Fig. 9 compares the accuracy of different research papers using EfficientNet. Each research paper had used either different or similar datasets and different classifiers, with specific values indicated on the vertical axis for accuracy. The most accurate case of EfficientNet had been with the NN classifier and RSNA dataset, which had achieved 86%.

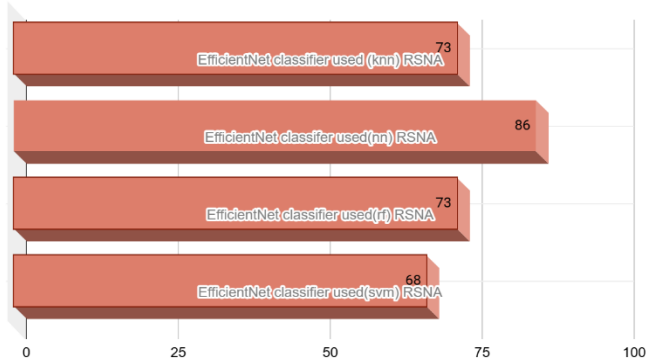


Fig. 9. Accuracy Vs Techniques using EfficientNet

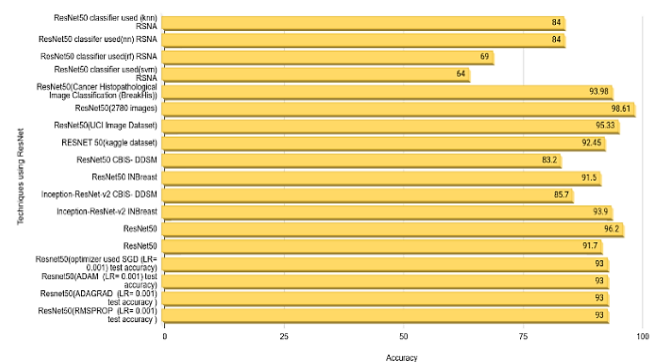


Fig. 10. Accuracy Vs Techniques using ResNet50

In Fig. 10 the highest accuracy that ResNet50 had accomplished was achieved by using a certain fixed given set of images. In this case, the highest value had been 98.61%. The maximum score that VGG16 had achieved had been using the BreakHis corpus, reaching the level of 98.14%.

One way to have used a custom CNN model to obtain the maximum accuracy had been by using a preset number of images that had reached 100% as shown in Fig 12.

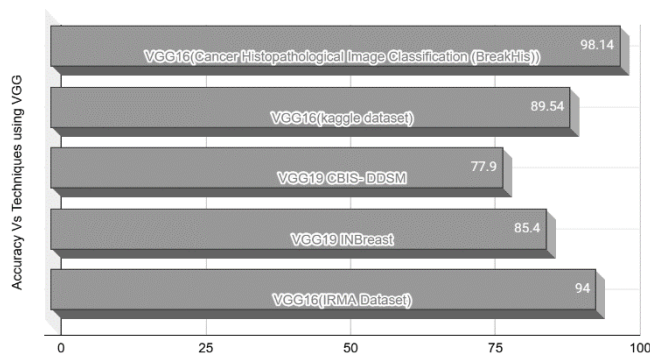


Fig. 11. Accuracy Vs Techniques using VGG16

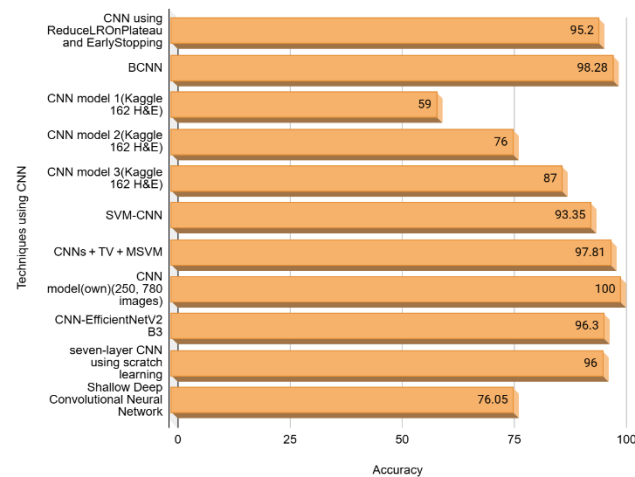


Fig. 12. Accuracy Vs Techniques using CNN

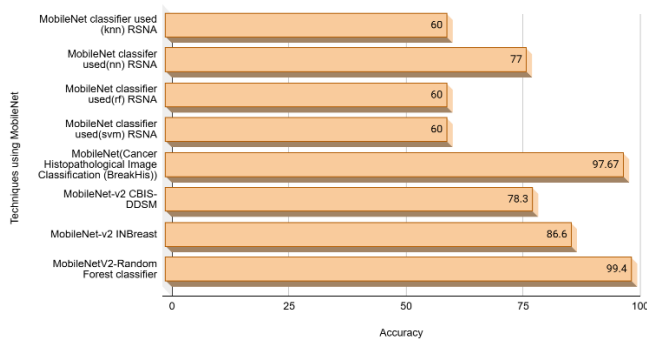


Fig. 13. Accuracy Vs Techniques using MobileNet

In Fig 13, Random forest classifier had given a maximum accuracy of 99.4% with MobileNetV2.

The highest accuracy that InceptionNet had achieved using the LPP feature enhancement was 99.81% as shown in Fig 14.

In Fig 15, the top result that Xception had achieved in classifying breast cancer was 97.85% using an image set.

The Concat model with the DDSM dataset had achieved the highest accuracy of 99.81% as shown in Fig 16.

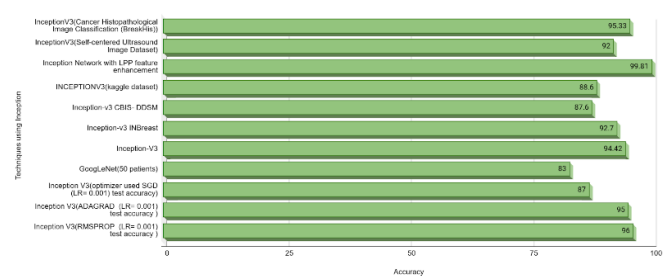


Fig. 14. Accuracy Vs Techniques using InceptionNet/GoogleNet

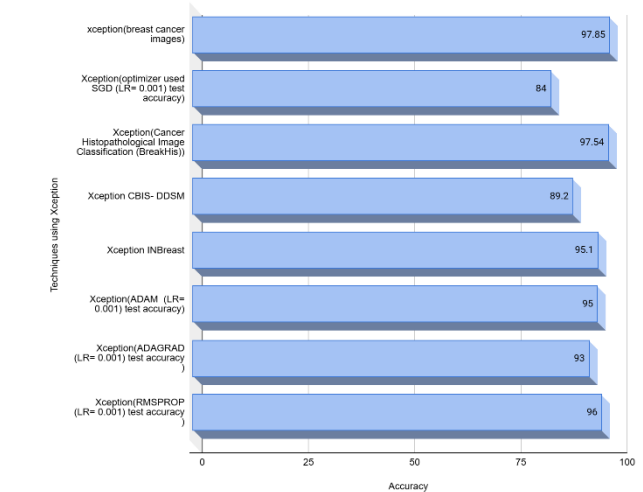


Fig. 15. Accuracy Vs Techniques using Xception

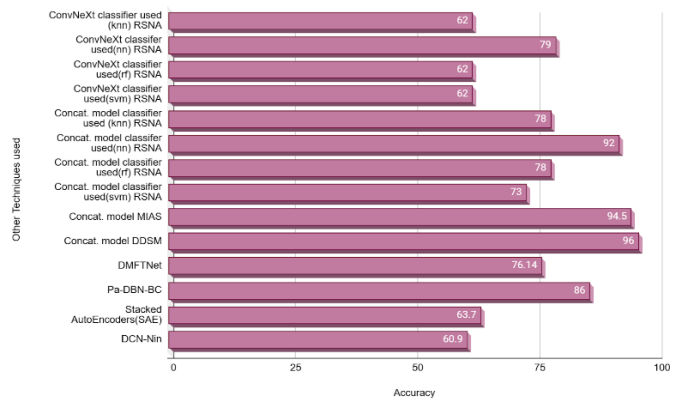


Fig. 16. Accuracy Vs Other techniques used

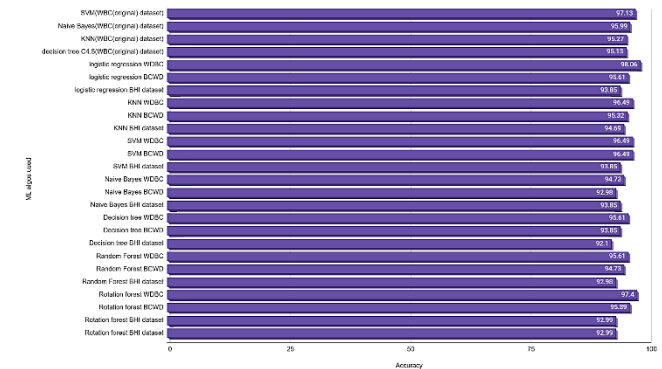


Fig. 17. Accuracy Vs using Machine Learning algorithms

E. Techniques using Machine Learning algorithms

Every figure has “Accuracy” on the x-axis and “Techniques Used” on the y-axis. In Fig. 17, the highest accuracy was achieved using the WDBC dataset using Logistic Regression, reaching 98.06%.

V. CONCLUSION

Breast cancer remains a formidable challenge due to the prevalence of improper and late diagnoses, underscoring the critical need for early and accurate detection methods. Recent advancements in machine learning and deep learning have significantly enhanced breast cancer classification, with models like InceptionNet achieving remarkable accuracies of up to 99.81%. However, the variability in model performance across different datasets highlights the necessity for more generalized models that can maintain high accuracy under diverse imaging conditions. This study emphasizes the pivotal role of deep learning networks, such as ResNet18 and ResNet34, in precise image classification. Combinations like MobileNet with Random Forest classifiers have shown success with minimal error, achieving 99.4% accuracy. Convolutional Neural Networks (CNNs) have demonstrated the potential for 100% accuracy in certain datasets, suggesting their clinical applicability. The surge in research publications in 2023 indicates a heightened focus on improving diagnostic methods. Despite these advancements, challenges persist. Models often excel in specific datasets but struggle in real-world applications, necessitating domain adaptation techniques to enhance generalizability. Deep learning models require substantial computational resources, complicating real-time analysis, which can be addressed by lightweight AI models and edge computing solutions. Additionally, dataset diversity due to demographic discrepancies can lead to inaccurate diagnoses, emphasizing the need for more representative datasets. Explainable AI (XAI) is crucial for transparency and trust in clinical settings. Future research should focus on developing adaptable models that incorporate feature enhancement techniques and smart ensemble approaches. Addressing these challenges will not only make AI-supported diagnostics more trustworthy but also enhance their interpretability and accessibility for global clinical applications. By integrating diverse data types, such as imaging, histopathology, and genomic data, diagnostics can become more comprehensive and effective. This holistic approach will be pivotal in revolutionizing breast cancer detection and treatment worldwide.

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